

SURVEY

A Taxonomy of Knowledge Bases for Retrieval-Augmented Methods in Vision: A Comprehensive Survey

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ABSTRACT Recent vision–language models (VLMs) demonstrate impressive performance but often hallucinate when essential knowledge is not encoded in their parameters. Retrieval-augmented generation (RAG), which integrates external knowledge bases (KBs), has thus been widely adopted in computer vision. Beyond VLMs, retrieval augmentation has also been applied independently to improve task performance. However, no systematic analysis has investigated which types of KBs are most suitable for particular vision tasks. In this study, we present the first comprehensive survey of more than sixty studies (2021–2025) with a novel focus on KB types. We propose a taxonomy consisting of six categories: unstructured text (V-UT), ontology (V-OT), image (V-IM), image–text pairs (V-IT), structured graphs (V-SG), and domain-specific data (V-DM). For each category, we review retrieval pipelines, downstream tasks, representative datasets, and indexing strategies under consistent criteria. Our analysis shows that, regardless of KB type, most systems converge on dense encoders with vector databases, reflecting a mature technical stack. Nevertheless, this convergence often underutilizes KB-specific structures, highlighting significant opportunities for future studies. Finally, we provide practical guidelines for KB selection and retrieval design in vision and emphasize the need for KB-specific retrieval methods and standardized benchmarks.

INDEX TERMS Computer vision, knowledge base, visual RAG, vision–language models (VLMs).

I. INTRODUCTION

In computer vision (CV), extensive efforts have aimed to improve model performance, with vision–language models (VLMs) achieving notable success in tasks such as visual question answering (VQA), image captioning, and long-form video understanding [1], [2], [3]. However, recent VLMs still suffer from hallucinations and omission of critical evidence, particularly when fine-grained details or up-to-date knowledge absent from parameters are required. For example, the

InstructBLIP model [4] has been reported to hallucinate in about 30% of responses, producing non-existent objects, inaccurate descriptions, or incorrect relationships. This issue is especially critical in real-world applications such as healthcare, autonomous driving, and remote sensing, where responses must be reliable and evidence-based. Addressing this challenge requires mechanisms that can integrate external information beyond training data.

Retrieval-augmented generation (RAG) addresses this issue by retrieving relevant information from an external knowledge base (KB) and integrating it into the input context of the generative model. Here, a KB refers to an external

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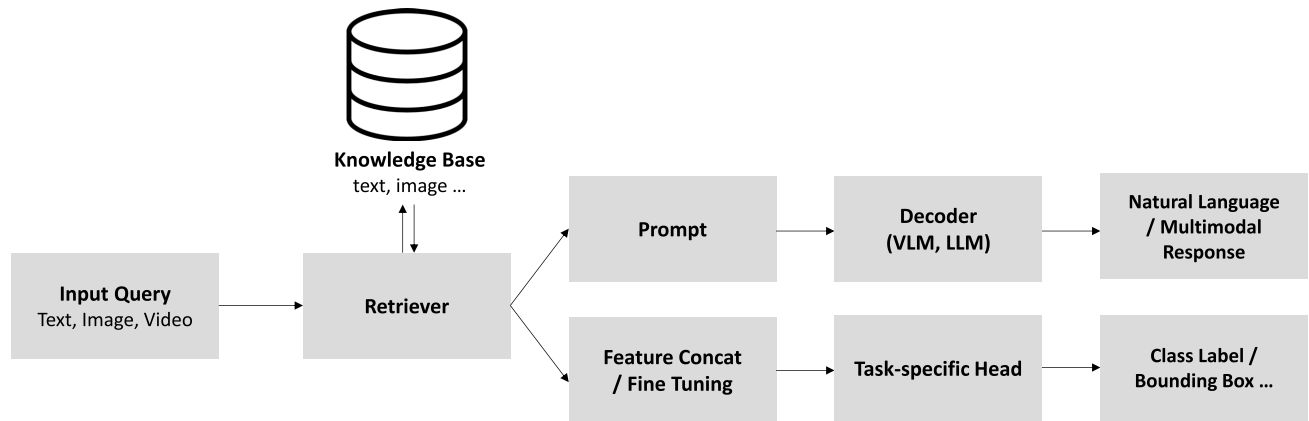


FIGURE 1. Two primary paths of retrieval-augmentation in computer vision.

collection of information accessible via queries beyond model parameters, ranging from structured knowledge graphs to large-scale unstructured datasets [5], [6]. The concept of RAG was formalized by Lewis et al. [7] and extended by Guu et al. [8], showing that incorporating retrieval during pre-training enhances generalization. Overall, the retriever fetches relevant evidence from large corpora such as Wikipedia, while the generator integrates the query and evidence to produce outputs. This approach offers a practical means of ensuring recency, accuracy, and evidential grounding, which fixed parameters alone cannot guarantee.

RAG has rapidly expanded in natural language processing (NLP) tasks such as domain-specific question answering (medicine, finance, education), text generation, summarization, and information extraction [9], and has more recently been introduced into the CV domain. In CV, external KBs are utilized through two main approaches. The first, the generation approach, inserts retrieved evidence (text, image, or multimodal) into prompts or decoder memory, followed by generation. Systems such as Visual-RAG [10] and EchoSight [11] exemplify this approach, leveraging up-to-date evidence in tasks requiring explanatory power, including VQA, video description, and report generation. The second, the understanding/adaptation approach, retrieves information and integrates it as features or restructures it into training data for adaptation. For instance, REACT [12] improved accuracy on diverse downstream tasks by selectively fine-tuning modules with retrieved web image-text pairs, while Domain-RAG [13] enhanced few-shot detector training efficiency by retrieving domain-consistent background images. The contrast between these approaches is illustrated in Fig. 1, which depicts only the principal processes. In practice, pipelines vary by task and data. Our proposed taxonomy adopts a KB-centric perspective that transcends these two approaches rather than depending on them.

Meanwhile, the spectrum of multimodal KBs has rapidly expanded, encompassing Wikipedia articles, ontologies, large-scale image-text pairs, scene graphs, and domain-specific data in areas such as healthcare, finance,

and remote sensing. However, few systematic studies in CV identify which KB types suit specific tasks and which indexing or retrieval strategies are most effective. Existing surveys primarily classify models by downstream tasks [2], [3], [6], [9], [14], [15], [16], making it difficult to compare pipelines under a unified lens of KB type, structure, and indexing method. Consequently, comparisons remain fragmented, new researchers lack a clear overview of suitable KBs for vision tasks, and deriving practical retrieval or indexing designs tailored to KB types is challenging. Our review identified few studies that systematically organize retrieval-augmented methods for CV while explicitly using KB type as the primary axis of classification. To address this gap, we aim to provide systematic guidance for vision researchers on selecting KB types and retrieval strategies, contextualizing existing work, and suggesting directions for future RAG applications in vision.

Overcoming these limitations requires clarifying the key factors driving RAG performance. At the core of RAG lies retrieval rather than generation. Specifically, KB selection, indexing, retrieval, and integration fundamentally determine model performance. Even with the same generative model, variations in KB type, representation unit, alignment strategy, indexing or retrieval method, and integration point (e.g., prompt vs. decoder memory) can yield substantial differences in quality, cost, and latency. Indeed, in RAG models such as Atlas [17], question answering (QA) accuracy varies significantly with document indexing despite identical generative architectures, underscoring that retrieval quality is the key determinant of overall RAG performance. This complexity is amplified in the vision domain. For example, in VQA, when queries involve fine-grained attributes or relationships, encyclopedia-style KBs such as Wikipedia [18] or ConceptNet [19] are effective [20], [21], [22]. Conversely, when up-to-date trends or entity-specific information are required, large-scale web image-text pairs retrieved from the Internet are more effective [12], [23]. Similar patterns emerge in domain-specific scenarios. Zhong et al. [24] constructed a RAG benchmark for chemistry and found that unstructured

text improved tasks such as molecular structure design and reaction prediction, whereas structured databases were more effective for nomenclature and terminology conversion. These findings highlight that the type of external data retrieved critically influences model performance.

In this study, we focus on the retrieval stage and systematize retrieval-augmented methods for vision from a KB-centric perspective, with emphasis on works published between 2021 and 2025. We classify the recurrently observed KB types into six categories and analyze each category by retrieval pipeline patterns, major downstream tasks, representative datasets, and indexing strategies within a unified framework. This classification is independent of whether retrieval is used for generation or understanding/adaptation; it focuses solely on the retrieval stage, though hybrid designs are possible in practice.

- 1) V-UT (Vision-Retrieval from Unstructured Text)
- 2) V-OT (Vision-Retrieval from Ontology Text)
- 3) V-IM (Vision-Retrieval from IMage), including frames extracted from videos
- 4) V-IT (Vision-Retrieval from Image Text pairs)
- 5) V-SG (Vision-Retrieval from Structured Graphs)
- 6) V-DM (Vision-Retrieval from DoMain-specific KBs)

The main contributions of this study are summarized as follows:

- We propose a KB-centric taxonomy that treats KBs as the core variable in vision retrieval tasks, classifying them into six categories: V-UT, V-OT, V-IM, V-IT, V-SG, and V-DM.
- We reorganize more than sixty recent studies under a unified framework, presenting datasets, benchmarks, and representative pipelines for each category to enable a comprehensive overview.
- We provide a comparative analysis of retrieval and indexing patterns, as well as evidence integration strategies across KB types, identifying practical design choices for specific vision tasks.

II. RELATED SURVEYS

Most existing RAG surveys focus on large language models (LLMs), summarizing core pipeline components and evaluation metrics, with discussions centered on applications in NLP. Gao et al. [6] defined three developmental stages—Naïve, advanced, and modular RAG—and systematically categorized technological trends for each. Arslan et al. [9] reorganized RAG works for LLMs by task and domain, highlighting domain-specific applications. More recently, studies have targeted multimodal RAG (MRAG). Mei et al. [14] compared text-based RAG and MRAG stage by stage, structured the MRAG study landscape into guidelines and roadmaps, and identified technical gaps and unresolved issues. They proposed a taxonomy of KBs with three categories: unstructured, semi-structured, and structured multimodal data. Abootorabi et al. [15] categorized MRAG by retrieval strategies and subdivided modality-centric retrieval

into text-centric, vision-centric, video-centric, audio-centric, and document-based approaches.

From the vision perspective, the most relevant prior work is by Zheng et al. [3], who categorized vision tasks along two axes—retrieval-augmented understanding and retrieval-augmented generation—introducing subcategories under each axis and surveying related studies. However, their survey organized tasks primarily by functional type, without distinguishing KBs, and did not provide KB-specific design patterns or dataset mappings. Liang et al. [2] surveyed applications of multimodal large language models (MLLMs) in vision, classifying them by task, but their focus was on MLLMs broadly rather than retrieval-augmented methods, diverging from this work. Also, Zhao et al. [16] classified MRAG by modality (image, code, structured knowledge, audio, video), yet did not adopt KB type or structure as the primary axis, nor provide pipeline designs, indexing strategies, or dataset mappings, and their discussion was not vision-centric. Despite these surveys, no prior work has classified retrieval-augmented methods in vision according to KB types.

III. A NEW TAXONOMY OF KNOWLEDGE BASES

In this section, we analyze prior studies by KB type, define six categories (V-UT, V-OT, V-IM, V-IT, V-SG, V-DM), and describe the characteristics of each category.

A. CLASSIFICATION CRITERIA

The core of RAG lies in retrieval, which searches for knowledge relevant to a query from an external KB. Recently, studies in CV have increasingly leveraged external knowledge to enhance image understanding and task performance. Our study of retrieval-augmented methods in vision identifies six major categories of external knowledge. Earlier works have attempted to classify RAG by knowledge type. To differentiate from prior work, we introduce a systematic categorization of how RAG is applied in vision tasks. Specifically, we define six KB types: unstructured text, ontology, images, image–text pairs, structured graphs, and domain-specific data. For each type, we highlight representative studies, analyze common characteristics, and examine major tasks and datasets.

B. DEFINITION AND OVERVIEW OF CATEGORIES

We outline the criteria for classifying the six categories and briefly summarize their representative characteristics. Detailed descriptions of corresponding studies, tasks, and datasets are provided in the following section.

1) V-UT (VISION-RETRIEVAL FROM UNSTRUCTURED TEXT)

V-UT refers to cases where unstructured text serves as external knowledge. In this approach, documents related to an image or query are retrieved to supplement knowledge absent from visual information. This resembles RAG in LLMs, where external knowledge typically consists of natural language passages. Such methods are widely applied to

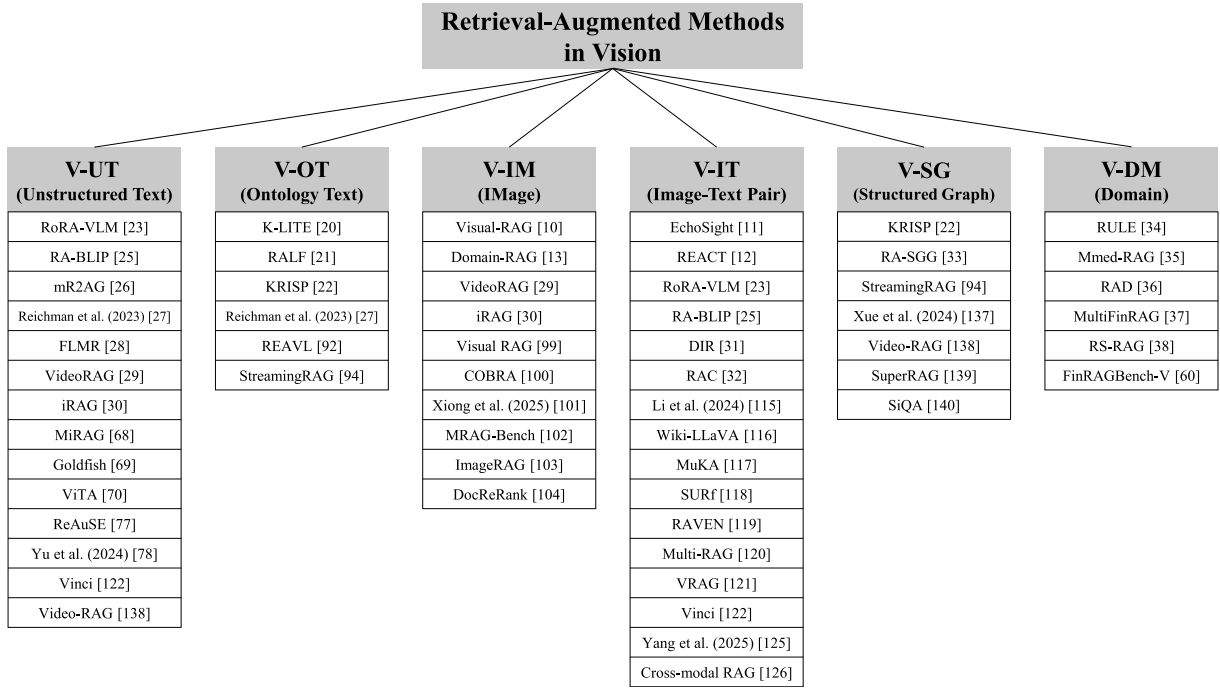


FIGURE 2. Taxonomy tree for retrieval-augmented method in vision.

knowledge-based VQA tasks requiring commonsense or background knowledge [25], [26]. They operate by retrieving text passages relevant to the query and incorporating them into answer generation. Typically, queries and candidate documents are embedded, similarity search retrieves the most relevant text, and the results are integrated into the LLM prompt for answer generation [27], [28].

2) V-OT (VISION-RETRIEVAL FROM ONTOLOGY TEXT)

V-OT refers to cases where structured text, such as ontologies, serves as external knowledge. By leveraging explicit hierarchies and definitions, models can reason about novel concepts unseen during training. Based on object labels in the query image, external data provide attributes or classification information. This approach has been applied to zero-shot image classification [20] and open-vocabulary object detection [21], enabling recognition of unseen classes and detection of categories beyond the training set.

3) V-IM (VISION-RETRIEVAL FROM IMAGE)

V-IM refers to cases where external images serve as KBs. Still images are most common, but videos can also be included after conversion into frames or clips. This category also includes cases where images only have class labels. Queries are embedded with visual encoders (e.g., CLIP), and similar images are retrieved via top- k cosine similarity. Retrieved images support tasks such as few-shot learning [13] or VQA [10], enhancing accuracy and synthesis quality. We do not introduce a separate Video KB type because our taxonomy is organized by the retrieval target representation rather than the input modality. In practice, video knowledge is typically decomposed into static frames or clips (V-IM)

and/or textual transcripts (V-UT). As a result, video-RAG pipelines are naturally treated as hybrid systems (V-IM+V-UT), as exemplified by VideoRAG [29] and iRAG [30].

4) V-IT (VISION-RETRIEVAL FROM IMAGE-TEXT PAIR)

V-IT refers to cases where external knowledge consists of image-text pairs. Each image-caption pair is treated as a unit, effective for tasks requiring both visual and textual context, such as image captioning [31] and long-tail classification [32]. Unlike V-IM, which may use images with only class labels, V-IT leverages rich descriptive captions that provide detailed visual and linguistic context. A query image is matched to captioned corpora, and retrieved captions or labels guide caption generation or classification. This enables the model to leverage broad visual-linguistic knowledge with limited training.

5) V-SG (VISION-RETRIEVAL FROM STRUCTURED GRAPH)

V-SG refers to cases where structured graphs (e.g., scene graphs or object relationships) serve as external knowledge. Unlike V-OT, which addresses semantic hierarchies, V-SG emphasizes spatial and interactional relationships among objects. Graphs may be constructed from images for query processing [33], or pre-existing knowledge graphs retrieved for reasoning [22]. This approach is particularly effective for fine-grained relational queries (e.g., object counts or relative positions).

6) V-DM (VISION-RETRIEVAL FROM DOMAIN-SPECIFIC KB)

V-DM refers to cases where domain-specific data used, such as in medicine [34], [35], autonomous driving [36], finance [37], and remote sensing [38]. Crucially, this

TABLE 1. Comparison of general-purpose applications on domain data vs. dedicated V-DM systems.

Aspect	General-Purpose App.	Dedicated V-DM Sys.
Scope	Same model/pipeline; only data changes.	Pipeline redesigned for the domain.
KB Source	None or generic; relies on parametric knowledge.	Explicit, curated domain KB (manuals, ontologies).
Retrieval	Generic similarity search; weak constraints.	Domain-aligned units; auditable evidence links.
Goal	Probe zero-shot generalization on tasks.	Reliable decisions under domain constraints.

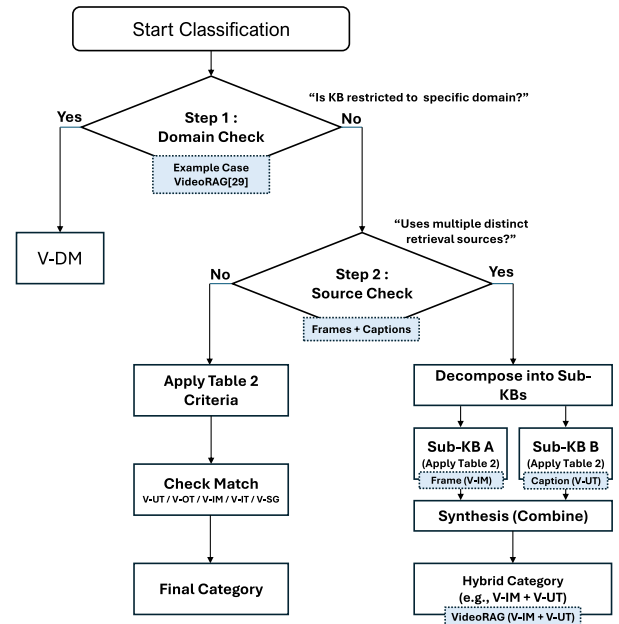
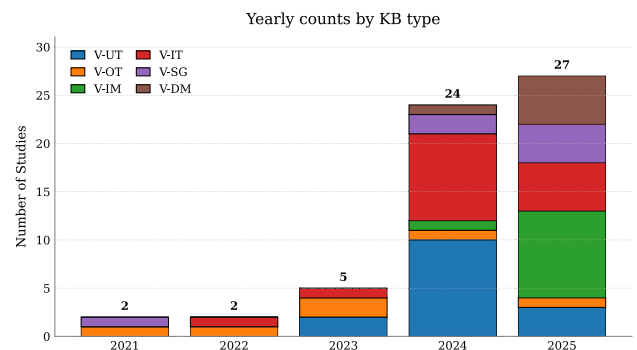
category is not defined by topic alone but by an ecosystem of constraints, including specialized or domain-pretrained encoders, domain-specific evaluation protocols (e.g., citation-grounded factuality and expert review consistency), and explicit requirements for safety, auditability, and provenance. These properties make V-DM qualitatively distinct from simply applying a general-purpose model to a new dataset: retrieval and evidence handling must satisfy compliance and risk controls in regulated settings, and typically enforce verifiable grounding even outside strict regulation. Domain knowledge is difficult to capture with general-purpose VLMs; thus, performance improves by retrieving and incorporating specialized literature, reports, or case data. Domain-specific encoders embed queries, retrieve similar past cases, and supply this knowledge to LLMs or VLMs [36], [38], making significant gains in expert-level tasks.

As summarized in Table 1, V-DM should be understood as an end-to-end domain-engineered ecosystem with an explicit domain KB, domain-aligned retrieval units, and auditable evidence links rather than a generic model evaluated on domain-shifted data.

C. TAXONOMY AND EVIDENCE FOR CATEGORY ASSIGNMENT

Fig. 3 provides the decision procedure for category assignment, and Table 2 summarizes the minimum requirements for each category. A study qualifies for a category only if it meets all items in that category's checklist. In particular, works that retrieve from multiple heterogeneous sources are classified as hybrid (e.g., V-IM + V-UT [29]), whereas intrinsically paired image-text corpora are assigned directly to V-IT (i.e., non-hybrid) rather than being split into separate V-IM and V-UT sources.

Fig. 2 presents a taxonomy tree summarizing the studies in each category. Some articles appear in multiple categories when different KB types are used separately. In contrast, if image and text are paired as unified KBs, the work is classified under V-IT. Fig. 4 summarizes the number of

**FIGURE 3.** Decision flowchart for KB classification. Prioritizing domain constraints (V-DM), it separates single-source from hybrid systems. Hybrid models (e.g., VideoRAG [29]) synthesize distinct components (e.g., V-IM + V-UT), unlike aligned V-IT pairs.**FIGURE 4.** Year-wise classification of studies covered in this survey (2021–2025.08).

papers published per year for each KB category and shows a steady upward trend. Notably, V-UT, V-IM, and V-IT constitute the largest shares, and the overall volume of studies continues to grow over time. The 2025 counts, however, include publications only up to August.

D. REPRESENTATIVE RETRIEVAL DATASETS

The datasets used in the retrieval stage for each category—that is, those serving as underlying knowledge bases—are listed in Table 3.

IV. CLASSIFICATION OF STUDIES BY PROPOSED TAXONOMY

In this section, we introduce studies corresponding to the six categories (V-UT, V-OT, V-IM, V-IT, V-SG, V-DM), describing retrieval pipeline patterns, major downstream tasks, datasets, and KB storage and indexing.

TABLE 2. Checklist for KB category assignment.

Category	Binary Checklist
V-UT	1) Knowledge is provided as natural-language passages or documents without explicit schemas/triples. 2) Retrieval uses text-based similarity (dense or sparse) over passages/documents. 3) Retrieved text is inserted directly into the LLM prompt, with no graph operations required.
V-OT	1) External knowledge is provided as an ontology or structured KB with explicit entities and relations (e.g., RDF/OWL/triples). 2) Queries resolve through entity/property identifiers or relation paths (e.g., SPARQL-like lookups). 3) Reasoning operates over the ontology's structure (hierarchies and relations), rather than free-text similarity.
V-IM	1) External knowledge consists of images that do not require paired captions for retrieval. 2) Retrieval is driven by visual embeddings (global or patch-level), rather than caption similarity. 3) Nearest-neighbor search over image features serves as the primary selection mechanism.
V-IT	1) External knowledge units are image–text pairs treated as first-class items. 2) Retrieval ranks candidates using cross-modal alignment between the query and the associated captions or labels. 3) Retrieved captions or labels are consumed directly to guide generation or classification.
V-SG	1) External knowledge is a structured scene or relationship graph of visual entities. 2) Retrieval or selection operates through graph matching, subgraph search, or path-based queries. 3) Answers rely on the object–relation structure, including counts and spatial or relational cues.
V-DM	1) External knowledge is confined to a specific domain (e.g., medical) and includes domain-specific metadata or protocols. 2) The domain constraint substantially governs retrieval and usage (e.g., institutional corpora, standardized codes). 3) Any internal structure (text, graph, image, or paired data) is secondary to the domain constraint.

TABLE 3. Representative retrieval datasets.

Category	Relevant Datasets
V-UT	Wikipedia 21M [18], Google Search GS112K, Google Search Corpus (168K) [39], InfoSeek 100K [40]
V-OT	Wikidata [41], WordNet [42], DBPedia [43], ConceptNet [19], Wiktionary [44], V3Det Vocabulary Store [45]
V-IM	ImageNet-1K [46], iNaturalist [47], CUB-200 [48], DTD [49], Oxford-IIIT Pets [50], TerraIncognita [51]
V-IT	COCO [52], LAION-400M / 5B [53], WIT [54], ShareGPT4v-PT [55], SBU-1M [56]
V-SG	AUG [57], GQA [58], VG-150 [59]
V-DM	FinRAGBench-V (Finance/Economics) [60], SEC's EDGAR filings (corpus) [61] (Finance/Economics), IU-Xray (Medical) [62], MIMIC-CXR (Medical) [63], PMC Open Access Subset (Medical) [64], [65], Quilt-1M (Medical) [66], RSWK (Remote Sensing) [38], NuScenes (Autonomous Driving) [67]

Although datasets for each category were outlined earlier, here we include both benchmark datasets used in experiments and datasets serving directly as knowledge bases.

A. V-UT (VISION-RETRIEVAL FROM UNSTRUCTURED TEXT)

1) RETRIEVAL PIPELINE PATTERNS

Retrieval pipeline designs in V-UT can be summarized into five major patterns.

First, dense embedding–based vector indexes have become the standard. Studies such as Reichman et al. [27], MiRAG [68], RA-BLIP [25], RoRA-VLM [23], Goldfish [69], ViTA [70], iRAG [30], and FLMR [28] embed queries and documents (or video clips) with encoders such as DPR [18], CLIP [71], or BERT [72], followed by retrieval using FAISS [73], HNSW [74], PLAID [75], or simple in-memory MIPS [76]. To align visual with textual features, many systems employ mapping networks, as in FLMR [28] and Reichman et al. [27].

Second, multi-stage or late-interaction retrieval strategies are widely used. MiRAG [68] and RoRA-VLM [23] employ two-step pipelines from ‘entity (or image) to passage’, while mR2AG [26] introduces Retrieval-Reflection and Relevance-Reflection tokens. FLMR [28] uses ColBERT-style token-level similarity scoring for fine-grained precision.

Third, several works integrate retrieval feedback into generation to improve end-to-end performance. ReAuSE [77] refines retrievers via reinforcement learning with QA rewards; mR2AG [26] re-ranks candidate answers with feedback; and iRAG [30] iteratively gathers evidence through a Planner–Extractor loop.

Fourth, some approaches directly project visual features into textual space. Reichman et al. [27] and RoRA-VLM [23] merge CLIP/ALIGN embeddings with language embeddings via averaging or linear mapping, while mR2AG [26] and FLMR [28] employ late-interaction mechanisms to precisely match multimodal queries with textual KBs. The goal

TABLE 4. V-UT major downstream tasks.

Category	Related Works
Knowledge-based VQA	ReAuSE [77], Reichman et al. [27], mR2AG [26], MiRAG [68], RA-BLIP [25], FLMR [28]
Visual entity identification / grounding	RoRA-VLM [23]
Relational / spatial description generation	Yu et al. [78]
Long-/short-form video QA	Goldfish [69], ViTA [70], iRAG [30]
Multimodal multi-hop QA / conversational QA	RA-BLIP [25]

TABLE 5. V-UT datasets.

Type	Representative Datasets
VQA Benchmarks	OKVQA [79], A-OKVQA [80], InfoSeek [40], Encyclopedic-VQA [81], ViQuAE [82], FVQA [83]
Entity/Passage KB	Wikipedia, Wiki21M [18], Google Search GS112K [39], Google Search Corpus (168K) [39], InfoSeek 100K [40]
Multimodal alignment/learning	WIT (37M full / 87K subset) [54]
Long-form video evaluation	TVQA-Long [69], MovieChat [84], Movie-QA [85], LLama-Vid [86], StreetAware [87], Tokyo MODI [88], MSR-VTT [89], MSVD [90], TGIF [91]

is to reduce the representational gap between vision and language, enabling query–document similarity within a unified space.

Finally, Wikipedia serves as the default KB in nearly all systems. Most works rely on Wikipedia 21M [18], supplemented with Google Search passages [39] or InfoSeek 100K [40]. Wikipedia’s broad topical coverage and open licensing make it the backbone KB for retrieval-augmented vision models.

2) MAJOR DOWNSTREAM TASKS

Downstream tasks for V-UT include knowledge-based VQA, entity identification/grounding, relational and spatial description, video QA (long- and short-form), multimodal multi-hop QA, and conversational QA. Question answering constitutes the majority. The related works are listed in Table 4.

3) COMMONLY USED DATASETS

Wikipedia-based corpora serve as the primary textual resources for V-UT. As summarized in Table 5, commonly used datasets include VQA benchmarks, entity/passage datasets, multimodal alignment datasets, and video-oriented datasets for long-form evaluation.

4) KNOWLEDGE BASE STORAGE AND INDEXING

The most widely used method is FAISS (Flat or HNSW), adopted by MiRAG [68], ViTA [70], and iRAG [30]. Reichman et al. [27] used a DPR-style index based on maximum inner product search (MIPS). FLMR [28] employed the PLAID late-interaction index, while ReAuSE [77] applied the Ferragina–Manzini (FM) index to generate and verify string-level identifiers via constrained decoding. Goldfish [69], given its small dataset, achieved real-time retrieval using in-memory cosine MIPS. iRAG [30] incorporated incremental updates: lightweight encoders first indexed the data, and during querying, the Extractor re-analyzed only relevant clips to append entries to the text database.

In summary, V-UT studies show that dense embedding–based vector retrieval has become the standard, typically implemented with FAISS/HNSW or PLAID [28], [30], [68], [70]. Multi-stage designs [23], [68] and enhancements such as Retrieval-Reflection or feedback mechanisms [26], [77] are widely used to improve precision. Wikipedia corpora serve as the primary KBs, often supplemented with Google Search passages [39] or InfoSeek data [40]. For downstream tasks, knowledge-based VQA is the primary application [25], [26], [27], [28], [68], [77], while video QA [30], [69], [70], entity grounding [23], and spatial relation description [78] have also been explored. Finally, although storage structures varied—from token-level representations (PLAID) [28], to substrings (FM-Index) [77], to vector databases [30], [68], [70]—all systems converged on MIPS or cosine similarity–based retrieval at the final stage.

B. V-OT (VISION-RETRIEVAL FROM ONTOLOGY TEXT)

1) RETRIEVAL PIPELINE PATTERNS

The common design patterns observed in V-OT can also be summarized into five patterns.

First, dense embedding–based vector indexes have become virtually standard. Reichman et al. [27] and REAVL [92] embed queries, images, and KB documents using encoders such as CLIP [71], ALIGN [93], or Bi-encoder [18], and then employ FAISS or general MIPS to retrieve top- k passages or entities in sublinear time.

Second, small-list in-memory cosine similarity search is used. RALF [21] stores the entire vocabulary or concept repository as an array in memory and directly computes cosine similarity between CLIP embeddings to select hard/easy negatives or augmentative concepts for augmentation. This lightweight approach avoids building a separate index while maintaining sufficient speed when the dataset size is relatively small.

Third, whole-graph loading with graph neural network (GNN) propagation appears as a design pattern. KRISP [22] and StreamingRAG [94] load triple-based graphs (e.g., DBpedia, ConceptNet, VisualGenome) entirely into GPU memory and apply relational graph convolutional networks (RGCN) or graph attention networks (GAT) to propagate messages and produce node-level logits. The

TABLE 6. V-OT major downstream tasks.

Category	Related Works
Knowledge-based VQA	KRISP [22], Reichman et al. [27], REAVL [92]
Image classification / Zero-/Few-shot	K-LITE [20]
Open-vocabulary object detection	K-LITE [20], RALF [21]
Real-time event / anomaly detection	StreamingRAG [94]
Multimodal entity linking	REAVL [92]

TABLE 7. V-OT datasets.

Type	Representative Datasets
Ontology / Lexicon-based	WordNet [42], Wikidata [41], DBPedia [43], ConceptNet [19], Wiktionary [44]
Encyclopedic / Passage	Wikipedia-21M [18]
Vocabulary lists	V3Det [45]
Visual graphs	VisualGenome [95]
Image-text pairs	WIT [54], YFCC-14M [96], GCC-12M [97], GCC-3M [98], COCO [52]

essence of this method is that relational inference is conducted directly through graph computation rather than separate indexing.

Fourth, lookup/definition-insertion prompts are employed. K-LITE [20] combines definitions and hierarchical paths retrieved from WordNet [42] or Wiktionary [44] with the original class name in the form of descriptive statements, which are then directly appended to the query. This inline knowledge integration into prompts enhances model performance without constructing dedicated vector indexes.

Finally, cross-modal alignment into a shared space is a recurring strategy. K-LITE [20], RALF [21], Reichman et al. [27], and REAVL [92] all leverage CLIP- or ALIGN-based dual encoders to align images, patches, and text/entities into the same embedding space. By directly matching multimodal query vectors with textual KB vectors, these systems minimize the representational gap between visual and textual knowledge.

2) MAJOR DOWNSTREAM TASKS

The downstream tasks addressed in V-OT include knowledge-based VQA, image classification, open-vocabulary object detection, real-time event or anomaly detection, and multimodal entity linking. Question Answering tasks are again the most dominant. The related works are listed in Table 6.

3) COMMONLY USED DATASETS

A variety of ontology- and lexicon-based datasets have been employed. Image and graph datasets are also used to complement ontology knowledge. In addition, many studies incorporate image-text pair datasets for pretraining. The relevant datasets are summarized in Table 7.

4) KNOWLEDGE BASE STORAGE AND INDEXING

FAISS is widely used, as in Reichman et al. [27] and REAVL [92]. Another approach performs direct cosine similarity using in-memory lists. RALF [21] applies this method, directly selecting from small vocabularies or concept sets without dedicated indexes. Finally, lookup tables are also used. K-LITE [20], for example, retrieves definitions and parent paths from WordNet and inserts them into prompts, enriching queries without constructing separate indexes.

In summary, V-OT studies show that dense vector representations with FAISS or MIPS are the most common [27], [92], though lightweight designs tailored to KB structure also exist, such as direct graph propagation [22], [94] or lookup-based prompt insertion [20]. Knowledge-based VQA remains the primary downstream task [22], [27], [92], with others including classification [20], open-vocabulary detection (OVD) [20], [21], and real-time event detection [94]. Core KBs include WordNet [42], Wikidata [41], and DBPedia [43], while datasets such as V3Det [45] are occasionally used for enrichment [21]. Finally, although storage methods vary-from vector databases to list-based cosine search, graph structures, and lookup tables-all approaches converge on cosine (inner-product) similarity at the final retrieval stage.

C. V-IM (VISION-RETRIEVAL FROM IMAGE)

1) RETRIEVAL PIPELINE PATTERNS

Retrieval pipeline designs in V-IM can be summarized into four main patterns.

First, dense embedding-based vector indexes are the de facto standard. Visual-RAG [10], Visual RAG [99], COBRA [100], Xiong et al. [101], iRAG [30], and MRAG-Bench [102] preprocess and vectorize inputs using encoders such as CLIP or ImageBind, then perform k -nearest neighbor retrieval from FAISS-based indexes, typically via cosine similarity or inner-product MIPS.

Second, some studies employ brute-force search. Domain-RAG [13], ImageRAG [103], Visual RAG (small-scale experiments) [99], and COBRA [100] (16 images per class) perform exhaustive comparison without approximate nearest neighbor (ANN) when KBs are small or latency tolerance is high.

Third, hybrid vision-text indexes are frequently adopted. VideoRAG [29], Xiong et al. [101], and iRAG [30] use both V-IM and V-UT data as KBs, performing parallel searches at query time and combining results through LLM-based intersection or filtering.

Finally, multi-stage retrieval with re-ranking is widely employed. Domain-RAG [13], VideoRAG [29], iRAG [30],

TABLE 8. V-IM major downstream tasks.

Category	Related Works
Few-shot learning	Domain-RAG [13], COBRA [100], Visual RAG [99]
Image-based visual knowledge QA	Visual-RAG [10], MRAG-Bench [102]
Long-form / streaming video QA	VideoRAG [29], iRAG [30], Xiong et al. [101]
T2I generation	ImageRAG [103]
Document page re-ranking	DocReRank [104]

TABLE 9. V-IM datasets.

Type	Representative Datasets
Image	COCO [52], LAION-2B [53], iNaturalist [47], ImageNet [46], CUB-200 [48], DTD [49], EuroSAT [106], Oxford Pets [50], TerraIncognita [51], HAM10000 [107], CheXpert [108], FIVES [109]
Video	LongerVideos [29], VQA-v2 [110], MSR-VTT [111], StreetAware [87], Tokyo MODI [88], StreamBench [101]
Domain / Document	ColPali 120K [112], ViDoReV2 [113], Real-MM-RAG [114]

and DocReRank [104] begin with coarse lightweight retrieval followed by fine-grained re-ranking using style distance, LLM signals, or ColBERT [105] late interaction.

2) MAJOR DOWNSTREAM TASKS

Downstream tasks include few-shot learning (classification, detection), image-based VQA, video QA, text-to-image (T2I) generation, and document re-ranking. The related works are listed in Table 8.

3) COMMONLY USED DATASETS

A wide range of image and video datasets have been used. Image datasets typically include captions or labels, while video datasets cover both short- and long-form collections. The relevant datasets are summarized in Table 9.

4) KNOWLEDGE BASE STORAGE AND INDEXING

As in other categories, FAISS is the most frequently adopted method. Visual-RAG [10], Visual RAG [99], COBRA [100], Xiong et al. [101], iRAG [30], and MRAG-Bench [102] all employed FAISS for efficient vector search. Brute-force cosine similarity search was also applied by Domain-RAG [13] and ImageRAG [103], which performed exhaustive comparisons without ANN due to small KB size or use in offline augmentation. VideoRAG [29], Xiong et al. [101], and iRAG [30] implemented hybrid graph-vector indexes

that processed entity graphs, text, and vision embeddings simultaneously. DocReRank [104] employed a multi-vector late-interaction approach, using dual token embeddings for more precise re-ranking.

In summary, V-IM studies typically adopt a default configuration combining dense CLIP embeddings with FAISS vector indexes [10], [30], [99], [100], [101], [102]. For large-scale KBs, most pipelines perform k -NN retrieval with this setup, while for small KBs or offline augmentation where latency is less critical, brute-force cosine/L2 comparisons are used as alternatives [13], [99], [100], [103]. Hybrid indexing structures that jointly process vision embeddings and textual cues have also emerged, enabling effective multimodal and long-context QA such as long-form or streaming video [29], [30], [101]. Retrieval designs further balance efficiency and accuracy with multi-stage strategies, combining lightweight coarse retrieval with fine-grained re-ranking [13], [29], [30], [104]. These methods support diverse downstream tasks, including few-shot learning [13], [99], [100], video QA (short and long form) [29], [30], [101], image-based VQA [10], [102], document re-ranking [104], and text-to-image (T2I) generation [103].

D. V-IT (VISION-RETRIEVAL FROM IMAGE-TEXT PAIR)

1) RETRIEVAL PIPELINE PATTERNS

Compared to other categories, V-IT includes the largest number of studies and correspondingly exhibits the greatest variety of datasets and retrieval designs. Common retrieval patterns can be summarized into five categories.

First, dense embeddings and vector indexes form the backbone of most systems. REACT [12], RAC [32], DIR [31], Li et al. [115], EchoSight [11], Wiki-LLaVA [116], RoRA-VLM [23], RA-BLIP [25], MuKA [117], SURf [118], RAVEN [119], Multi-RAG [120], VRAG [121], and Vinci [122] all rely on encoders such as CLIP, BLIP [123], or ViT [124] to vectorize images (or image-text pairs), followed by top- k candidate retrieval using FAISS or MIPS-based approximate k -NN search.

Second, hierarchical and multi-stage retrieval strategies are frequently used to balance efficiency and precision. Similar to other categories, RoRA-VLM [23], Wiki-LLaVA [116], VRAG [121], and Multi-RAG [120] perform a first-stage coarse search over anchor images or entities, followed by a second-stage fine-grained search over passages, shots, or text units. This approach reduces computational cost while retrieving necessary details.

Third, many systems include re-ranking or filtering stages instead of directly using initial retrieval results. VRAG [121], Yang et al. [125], MuKA [117], RA-BLIP [25], and SURf [118] employ scoring from large vision-language models, masking, or other strategies to re-order initial candidates or filter noisy entries, thereby improving final performance.

Fourth, studies such as MuKA [117], Yang et al. [125], and Reichman et al. [126] use token- or patch-level late

TABLE 10. V-IT major downstream tasks.

Category	Related Studies
Knowledge-Based VQA	EchoSight [11], Wiki-LLaVA [116], RoRA-VLM [23], MuKA [117], SURf [118], RAVEN [119], Yang et al. [125]
Video QA / Video Understanding	Multi-RAG [120], VRAG [121], Vinci [122]
Image Captioning	DIR [31], Li et al. [115], SURf [118], RAVEN [119]
Image Classification	REACT [12], RAC [32], SURf [118]
Image-Text Retrieval	REACT [12], RAVEN [119]
Entity Grounding	RoRA-VLM [23]
Multi-hop / Conversational QA	RA-BLIP [25]

TABLE 11. V-IT datasets.

Type	Key Datasets
Web-scale Image-Text Pairs	LAION-400M / 5B [53], WIT 37M [54], ShareGPT4v-PT 1.2M [55], SBU-1M [56], Web-800M [12]
Encyclopedic / Information-Seeking VQA	InfoSeek [40], Encyclopedic-VQA [81], OK-VQA [79], ScienceQA [127], MMMU [128], MathV [129]
General Vision Benchmarks	COCO [52], Flickr30K [130], NoCaps [131], VizWiz [132], ImageNet-1K [46], CIFAR-10 [133]
Video / Scene Corpora	HowTo100M [134], MMBench-Video [135], Ego4D [136]

interaction or multi-vector representations, storing documents or queries as multiple vectors and computing similarity via max-sum aggregation across tokens.

Finally, dynamic web or memory augmentation is also adopted in some works to supplement static KBs with real-time information. RoRA-VLM [23] augments first-stage entity retrieval with Google Search results in real time, while Vinci [122] maintains a memory DB of streaming scene-description logs, enabling immediate incorporation of past events at query time. These designs improve adaptability to dynamic or evolving scenarios.

2) MAJOR DOWNSTREAM TASKS

Among the downstream tasks, knowledge-based VQA on static images is the most prevalent. In addition, tasks commonly studied in CV-such as video QA, image captioning, and image classification-are also widely included. The related studies are listed in Table 10.

3) COMMONLY USED DATASETS

In V-IT, large-scale image-text pair datasets are predominantly used, supplemented by encyclopedic and

information-seeking VQA datasets to enrich textual content. In addition, various datasets and benchmarks for video processing have also been adopted. The relevant datasets are summarized in Table 11.

4) KNOWLEDGE BASE STORAGE AND INDEXING

As in other categories, FAISS is widely adopted. REACT [12], RAC [32], EchoSight [11], Wiki-LLaVA [116], RoRA-VLM [23], SURf [118], RAVEN [119], and Vinci [122] all employed FAISS for k -NN search over image or multimodal vectors. DIR [31] used an internal MIPS-based approximate k -NN, while RA-BLIP [25] and MuKA [117] also relied on MIPS as their indexing mechanism. Because image-text pair datasets are often large, HNSW graphs are frequently used for efficient large-scale indexing. Other storage methods also appear: Multi-RAG [120] used the Chroma Vector DB, while Yang et al. [125], MuKA [117], and Reichman et al. [126] used Dense MIPS, performing top- k retrieval directly through internal model multiplications for small KBs without external libraries. VRAG [121] and partially MuKA [117] adopted late-fusion or multi-vector indexing, storing multiple vectors per document or shot and re-ranking them at the patch or token level.

In summary, V-IT studies show that converting queries and KBs into dense embeddings and performing approximate k -NN search via FAISS has become the default design. In particular, HNSW variants allow indices with tens or hundreds of millions of entries to be processed in real time, making this approach highly effective for large-scale multimodal KBs [12], [23], [32], [118], [122]. Beyond single-stage retrieval, many systems further improve precision and efficiency with hierarchical retrieval or MLLM-based re-ranking [23], [116], [121]. Core KBs include COCO [52], LAION [53], and WIT [54], often supplemented with multimodal corpora such as HowTo100M [134] or ShareGPT4v [55]. For downstream tasks, knowledge-based VQA is dominant, but methods are also applied to video QA, long-tail image classification, image captioning, and other tasks. Finally, despite differences in storage structures, all systems converge on cosine/MIPS-based k -NN retrieval, with approximate nearest-neighbor search consistently forming the final stage.

E. V-SG (VISION-RETRIEVAL FROM STRUCTURED GRAPH)

1) RETRIEVAL PIPELINE PATTERNS

The design patterns commonly observed in V-SG can be summarized into four categories.

First, as in other types, dense embedding-based vector indexing has been most prominently employed. Xue et al. [137], Video-RAG [138], SuperRAG [139], and SiQA [140] embed representations obtained from scene graphs, optical character recognition (OCR) and automatic speech recognition (ASR) using SentenceBERT [141], Contriever [142], or Text2Vec [143], storing them in FAISS, Milvus, or in-memory indexes, and subsequently retrieving relevant information via cosine or MIPS-based Top- k search.

TABLE 12. V-SG major downstream tasks.

Category	Related Studies
Knowledge-Base VQA	Xue et al. [137], KRISP [22]
Video VQA	Video-RAG [138]
Scene Graph Generation & Detection	RA-SGG [33]
Real-Time Event Detection and Alarm	StreamingRAG [94]
Document / Layout QA	SuperRAG [139]
Structured Image QA	SiQA [140]

TABLE 13. V-SG datasets.

Type	Representative Datasets
Scene Graph	VG-150 [59], GQA [58], AUG [57]
Knowledge VQA	OK-VQA [79]
Long-Form Video Understanding	Video-MME [146], MLVU [147], LongVideoBench [148]
Document / Layout	DOCBENCH [149]
Structured Images	SCQA / OCQA (Organizational Charts) [140]

Second, graph-centric search has also been applied. StreamingRAG [94] and SuperRAG [139] store visual triples or layouts in symbolic knowledge graphs (KGs) such as Neo4J [144] or NetworkX [145], retrieving relevant subgraphs through graph traversal. In contrast, KRISP [22] directly extracted query-relevant subgraphs or node sets across the entire symbolic KG via RGCN-based global propagation.

Third, numerous cases adopt multi-source integration strategies for knowledge bases. KRISP [22], StreamingRAG [94], Video-RAG [138], and SuperRAG [139] simultaneously query structured visual graphs (V-SG) and unstructured corpora (V-UT), thereby enriching the LLM input context.

Finally, in scenarios requiring real-time or streaming updates, such as StreamingRAG [94], objects and relations are extracted frame by frame from images and incrementally inserted into spatiotemporal KGs. These elements are immediately reflected in queries, enabling effective real-time situational awareness.

2) MAJOR DOWNSTREAM TASKS

Regarding downstream tasks, knowledge-base VQA over static images was the most prevalent. In addition, a wide range of downstream tasks were conducted based on structured graph architectures, including VQA, scene graph

generation, and structured image QA. A summary of the related works is provided in Table 12.

3) COMMONLY USED DATASETS

For V-SG, numerous structured-graph datasets have been employed, along with benchmarks for evaluating vision tasks involving structured graphs. Some datasets also treat documents as structured images to support tasks in this domain. The relevant datasets are summarized in Table 13.

4) KNOWLEDGE BASE STORAGE AND INDEXING

As in other categories, FAISS is the most frequently used method, adopted by Video-RAG [138] and SiQA [140]. Xue et al. [137] used a temporary vector database, though the specific library was not disclosed. SuperRAG [139] combined Milvus and Elasticsearch to support graph traversal, while SuperRAG [139] and StreamingRAG [94] used Neo4J and NetworkX graphs, enabling queries over nodes, edges, and layouts via Cypher or Python APIs. KRISP [22] computed over its entire symbolic KG, applying RGCN-based global propagation before using node scores. RA-SGG [33] stored fewer than ten relation embeddings in an in-memory list and applied cosine k -NN linear search.

In summary, V-SG studies show that dense embedding-based vector search forms the backbone, with representations from scene graphs, OCR, or ASR embedded for top- k retrieval via FAISS, Milvus, or MIPS [137], [138], [139], [140]. In some cases, lightweight in-memory linear search is used instead, depending on dataset scale and latency requirements [33]. Beyond vector matching, a distinct trend is applying graph propagation and graph traversal to extract query-relevant subgraphs or node sets [22], [94], [139]. Hybrid retrieval combining structured visual graphs (V-SG) with unstructured text (V-UT) has also become common practice [22], [94], [138], [139]. For downstream tasks, knowledge-based VQA remains the central focus [22], [137], with the scope expanded to include scene graph generation [33], real-time event detection [94], document QA [139], and structured image QA [140].

F. V-DM (VISION-RETRIEVAL FROM DOMAIN-SPECIFIC KB)

1) RETRIEVAL PIPELINE PATTERNS

Unlike other categories, V-DM is classified separately due to the specialized nature of its KBs, and its retrieval patterns differ from previous cases.

First, dense embedding-based vector indexes remain consistent with the general trend. With larger KBs, vector databases for ANN search are employed. RS-RAG [38] used Qdrant collections, MultiFinRAG [37] adopted FAISS indexes, and RAD [36], while not naming the library, also stored embeddings in a vector DB for ANN search.

Second, RULE [34], MMed-RAG [35], RS-RAG [38], MultiFinRAG [37], FinRAGBench-V [60], and RAD [36] employed dense dual encoders to embed images and text (or two image types) separately, selecting top- k candidates via

TABLE 14. V-DM domains and major tasks.

Domain	Related Studies
Medical	Medical VQA, medical report generation (RULE [34], MMed-RAG [35])
Satellite Imagery	Image captioning, place classification, VQA (RS-RAG [38])
Finance	Multimodal financial QA (tables, charts, text, images), visual citation evaluation (MultiFinRAG [37], FinRAGBench-V [60])
Autonomous Driving	High-level action decisions (e.g., speed, lane change) (RAD [36])

TABLE 15. V-DM datasets.

Domain	Representative Datasets
Medical	MIMIC-CXR [63], IU-Xray [62], PMC-OA [65], Quilt-1M [66]
Remote Sensing	RSWK [38]
Finance	SEC's EDGAR 10-K/10-Q/8-K/DEF 14A PDFs [61], FinRAGBench-V [60]
Autonomous Driving	NuScenes forward-view and BEV images [67]

cosine or inner-product similarity. In other words, narrowing first-stage candidates via distance computation between query and KB vectors has been the most widely used strategy.

Third, modality-specific index separation is also employed. RS-RAG [38] stored image and text embeddings in distinct Qdrant collections, while MultiFinRAG [37] built separate FAISS indexes for text, tables, and images, applying thresholds sequentially at each stage. FinRAGBench-V [60] likewise maintained separate indexes for page images and OCR text. These separate indexes are later used for re-ranking or context fusion.

Fourth, for relatively small KBs, memory matrices are used instead of vector DBs. In the medical domain, RULE [34] and MMed-RAG [35] stored embedding matrices in GPU/CPU memory and extracted top- k results directly from the similarity matrix.

Finally, retriever pipelines were also tailored to domain-specific characteristics. MMed-RAG [35] routed input images into radiology or ophthalmology pipelines to apply domain-specific encoders. MultiFinRAG [37] applied stage-wise thresholds across modalities in the order text $\rightarrow$ tables $\rightarrow$ images. RAD [36] encoded both forward-facing and bird's-eye view (BEV) images from autonomous driving scenes into a single vector, enabling more precise similarity computation for driving scenes.

2) MAJOR DOWNSTREAM TASKS

Unlike other categories, V-DM studies divide tasks more finely by domain. Papers are therefore grouped by domain,

with specific tasks identified within each. The related works are listed in Table 14.

3) COMMONLY USED DATASETS

Since the studies corresponding to V-DM incorporate diverse datasets tailored to their respective domains, the datasets are organized by domain rather than by specific task. The relevant datasets are listed in Table 15.

4) KNOWLEDGE BASE STORAGE AND INDEXING

RULE [34] and MMed-RAG [35] extracted top- k results directly from in-memory matrices, suitable for rapid experimentation and fine-tuning given the relatively small KB size. RS-RAG [38] used Qdrant collections, separating large-scale image-text pairs and conducting cosine-based ANN search. MultiFinRAG [37] employed FAISS to store text, tables, and images separately by modality, while FinRAGBench-V [60] used a vector database (DB) mapping page images and OCR vectors to shared identifications. RAD [36] concatenated forward-view and BEV vectors to retrieve the top-1 scene using a single vector DB.

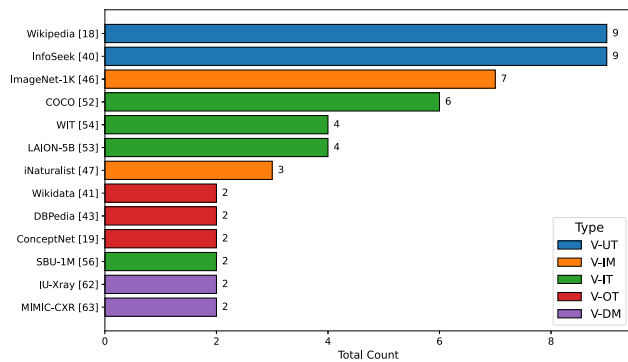
In summary, V-DM studies predominantly embed queries and KB entries using dense dual encoders, with top- k candidates selected via cosine or inner-product similarity [34], [35], [36], [37], [38], [60]. Depending on KB scale and modality complexity, two strategies emerge: (1) for smaller KBs in domains such as medicine, embedding matrices are stored in memory to compute top- k results directly [34], [35]; (2) for large-scale corpora in remote sensing, finance, and autonomous driving, vector DBs such as Qdrant and FAISS are used for ANN retrieval [36], [37], [38], [60]. The addressed tasks focus on practical applications, including medical VQA and report generation [34], [35], satellite image analysis [38], financial document QA [37], [60], and autonomous driving decision-making [36]. Storage and indexing strategies vary: some systems store all modalities in a single collection [36], while others separate modalities [37], [38], [60] to support re-ranking or context fusion. Nevertheless, two consistent approaches emerge across the V-DM literature: maintaining image-text mappings via shared IDs and performing cosine-based approximate search.

V. DISCUSSION

This study analyzed retrieval-augmented methods applied to vision tasks, focusing on works published in the past three years. While NLP retrieval (especially for LLMs) primarily relies on text-based KBs, the vision domain reveals a broader landscape, employing diverse KB forms. We categorized KBs into six types-V-UT (Unstructured Text), V-OT (Ontology Text), V-IM (Image), V-IT (Image-Text Pair), V-SG (Structured Graph), and V-DM (Domain-specific)-and examined pipeline design, downstream tasks, and datasets for each. We synthesize the core representations, retrieval and indexing principles, and downstream tasks for these six categories in Table 16.

TABLE 16. KB categories: representation, retrieval, and tasks.

Category	Knowledge Representation	Typical Retrieval / Indexing	Main Downstream Tasks
V-UT	Unstructured text, passages	Dense dual encoders; ANN (FAISS/HNSW); late interaction; PRF/query expansion	KB-VQA; entity grounding; video QA
V-OT	Ontology/structured text, definitions	KG traversal (SPARQL/Gremlin); KG-emb (GNN/RGCN); lookup prompts; cosine rerank	KB-VQA; zero/few-shot cls; OVD; entity linking
V-IM	Images/patches (incl. video frames)	CLIP-like encoders; brute-force (small) / ANN (large); hybrid text+image; 2-stage rerank	Few-shot cls/det; image-VQA; video QA; doc rerank
V-IT	Aligned image–text pairs	Dual encoders; HNSW; hierarchical/two-stage; late interaction; VLM rerank	KB-VQA; captioning; long-tail cls; image–text retrieval; grounding
V-SG	Visual graphs (scene/OCR/layout)	Vector+graph hybrid; graph propagation (RGCN); subgraph match; V-UT hybrid	KB-VQA; scene-graph gen; doc/structured-image QA; event detection
V-DM	Domain-specific multimodal KBs	Domain-tuned encoders; modality-silo indexes; ANN (FAISS/Qdrant); routing; memory matrices	Domain VQA; report gen; RS caption/cls/VQA; AD decision support

**FIGURE 5.** Frequency of external knowledge bases adopted as retrieval targets in our study: The chart counts how often each dataset serves as a retrieval knowledge source, rather than merely as an evaluation benchmark.

To analyze the trends in KB usage across the surveyed literature, we aggregated the frequency with which each dataset was used as a retrieval target (excluding evaluation-only usage). As shown in Fig. 5, general-purpose textual corpora such as Wikipedia (V-UT) and image–text corpora such as InfoSeek (V-UT) are the most frequently used KBs, each referenced in nine papers. These are followed by foundational vision and vision–language datasets such as ImageNet (V-IM) (7 papers) and COCO (V-IT) (6 papers).

A. COMMON CHARACTERISTICS OBSERVED

Across V-UT to V-DM, most systems converge on dense encoders with vector databases as the default retrieval backbone. Dense dual encoders with FAISS are common, and are often combined with late interaction, re-ranking, or multi-step retrieval. Structure-aware retrieval remains rare; intrinsic structures in V-SG (graph topology) and V-OT (hierarchical relations) are seldom leveraged beyond the embedding. Regarding downstream tasks, VQA is the predominant application across KB types, with retrieval augmentation also used for captioning, image and video classification, and video QA. Task distributions appear similar across

V-UT, V-OT, V-IM, V-IT, V-SG, and V-DM, suggesting that KB type does not align closely with any specific task family. Retrieval is typically evaluated using Recall@K, nDCG [150], or mean reciprocal rank (MRR) followed by task-specific metrics for generation or prediction. Variations in evidence quality, pipeline design, and computational or latency budgets further complicate direct comparisons across tasks. Overall, these trends highlight a clear research gap: the need for KB-specific, structure-aware retrieval methods that extend beyond generic dense encoders.

To contextualize this gap for V-SG and V-OT, we highlight recent operator-level advances that explicitly exploit graph topology and ontology structure during retrieval, rather than treating structure only as post-hoc reasoning input. Importantly, the key missing piece is not additional vision examples but a clearer account of structure-aware retrieval operators that move beyond flat top- k matching over linearized triples. Recent work makes this operator-level perspective explicit by introducing *i*) subgraph construction via parallel triple scoring with explicit structural features (e.g., SubgraphRAG [151]), *ii*) graph-guided expansion and KG-based context organization that traverses relations to diversify and arrange evidence (e.g., KG²RAG [152]), and *iii*) ontology-grounded hypergraph retrieval where evidence units are mapped to domain concepts/relations for verifiable grounding (e.g., OG-RAG [153]). These operators directly map to V-SG/V-OT settings, where scene-graph topology and ontology hierarchies can define the appropriate evidence unit (relation-preserving subgraphs), admissible expansions (typed multi-hop neighborhoods), and auditable verification interfaces (links to concepts/relations), strengthening grounding beyond treating graphs as mere sequences of triples.

B. INCREASING USE OF TWO KBs

Two forms of collaboration emerge when combining distinct KB types. Staged-pipeline collaboration performs sequential retrieval with role separation, such as candidate generation

TABLE 17. Comparison of representative multimodal RAG benchmarks. Recent benchmarks (FinRAGBench-V, REAL-MM-RAG) address domain-specific needs and real-world robustness compared to the vision-centric MRAG-Bench.

Benchmark	Domain	KB / Corpus Type	Scale	Key Metrics
MRAG-Bench [102]	General (Vision-centric)	- Image corpus 9 vision-centric scenarios (e.g., occlusion, viewpoint)	16,130 Images 1,353 MCQA	Retrieval: Recall@5 QA: Accuracy Analysis: Image-vs-Text Improv. (%)
FinRAGBench-V [60]	Finance (En/Zh)	- Financial PDFs - Page/Block citation labels	111,999 Pages 1,394 QA Pairs	Retrieval: nDCG, Recall, MRR QA: ROUGE, GPT-4o Acc Citation: Page/Block Prec. & Rec.
REAL-MM-RAG [114]	Real-World (Fin + Tech)	- Layout-heavy docs (reports, slides, manuals) - Query rephrasing tests	163 Docs (8,604pgs) 4,553 Queries	Primary: NDCG@5 Robustness: Performance drop under query rephrasing

followed by verification or reranking. Reported instances include V-IM—first candidate generation with V-UT verification [30], as well as V-OT filtering prior to dense reranking [22], [27], [94]; additional staged pairings include V-IM with V-UT [122] and V-IT with V-UT [23], [25].

Fused-KB integration jointly constructs indexing or retrieval signals from multiple KB types. At the indexing or representation layer, fusion is implemented through shared cross-V-* keys, joint embedding spaces, or co-index structures (when available). At the retrieval-scoring layer, structure-aware scoring has been reported for V-SG via graph-neural reranking and subgraph constraints [22], [94], [138] and for V-OT via symbolic path constraints and definition lookups [22], [27], [94]. In document-heavy V-UT pipelines, layout-aware late interaction is used as a scoring mechanism rather than an indexing mechanism [23], [25].

These hybrid designs also introduce challenges, including increased pipeline complexity, coordination of heterogeneous encoders, and potential conflicts between signals from different KBs – issues that future work will need to address.

C. FUTURE STUDIES

1) BENCHMARKS AND PROTOCOLS

As summarized in Table 17, MRAG-Bench [102] provides a vision-centric testbed for evaluating retrieval performance alongside downstream multiple-choice question answering (MCQA) accuracy. More recent benchmarks, such as FinRAGBench-V [60] and REAL-MM-RAG [114], extend the scope to domain-specific document corpora and robustness tests. Despite this progress, benchmarking efforts remain fragmented: evaluation protocols are non-uniform, and reproducibility is not consistently supported due to the uneven availability of public index-build scripts and fully specified indexing parameters. Moreover, system-level efficiency metrics such as latency, throughput, and memory footprint are rarely reported in a standardized form.

To move beyond disparate datasets and ad-hoc protocols, the community would benefit from a unified benchmarking framework. Such a framework should define minimum

requirements that any future benchmark—whether targeting V-UT, V-OT, V-IM, V-IT, V-SG, or V-DM—must satisfy to enable fair comparisons: *i*) fixed corpora accompanied by public index-build scripts and exact index parameters; *ii*) standardized retrieval metrics (e.g., Recall@K, nDCG) and answer-quality metrics (e.g., model-judged accuracy), evaluated under identical settings, together with robustness to query variations; *iii*) standardized reporting of system costs, including indexing time, query latency, throughput, and memory footprint; *iv*) auditable evidence-linking mechanisms (e.g., page/block/region-level grounding) that connect retrieved items to model outputs; and *v*) comprehensive reproducibility artifacts (e.g., random seeds, hardware specs, software versions).

Finally, this framework should provide reference implementations for multi-source pipelines and a schema to declare cross-V-type configurations, enabling controlled, like-for-like comparisons across diverse KB types and retrieval strategies.

2) RETRIEVAL-GENERATION INTERACTION

Beyond one-way retrieve-then-generate pipelines, interaction loops that use generator signals—such as reranker scores or answer confidence estimates—to adapt retrieval, or lightweight adapters that better align retriever and generator objectives, are likely to improve grounding and reduce hallucinations.

Recent case studies show that retrieval-generation feedback can yield measurable gains. For video-centric RAG, iRAG [30] triggers query-aware, on-demand extraction only when indexed information is insufficient, achieving 23-25× faster ingestion than exhaustive baselines; its filtering also improves retrieval efficiency on VQA-v2 (Recall@8: 66.07%→74.14%, avg. chunks: 6.88).

For knowledge-based VQA, mR2AG [26] tightens coupling via Retrieval-Reflection, Relevance-Reflection, and confidence-aware aggregation, improving accuracy by +10.6% (INFOSEEK-Human) and +15.5% (INFOSEEK-Wikidata) over prior baselines.

Accordingly, reporting should specify interaction details (trigger criteria, loop iterations) and system overheads (e.g.,

incremental extraction cost, latency budgets) to contextualize quality gains.

3) SCALABILITY AND REAL-TIME RETRIEVAL

At million-to-billion scale, systems must balance index build time, memory footprint, and latency, particularly when comparing generic dense vector databases with more specialized, structure-aware search schemes for certain KB types. Promising directions include IVF-PQ or HNSW hybrids, incremental indexing, cache-then-rerank pipelines, hot and cold shard separation, and streaming-friendly retrieval for temporal KBs, each evaluated under explicit latency and throughput budgets.

4) BEYOND STATIC KBs: FLUID AND GENERATIVE KNOWLEDGE SOURCES

While our taxonomy categorizes KBs by the retrieval target representation (e.g., V-IM, V-UT), static vs. fluid knowledge acquisition is an orthogonal axis: evidence may be retrieved from a fixed offline index or obtained dynamically at inference time. Recent trends increasingly consider such “fluid” settings, where evidence is acquired through interaction, tools, or model-driven query reformulation.

A representative example is WebArena [154], which frames web interaction in a realistic reproducible environment, allowing agents to perform actions and retrieve information from evolving webpages and embedded resources rather than a single frozen corpus. In parallel, recent frameworks integrate adaptive retrieval and generation. SELF-RAG [155] trains models to retrieve on-demand and critique retrieved evidence via reflection tokens, while ImageRAG [103] applies dynamic retrieval to visual generation by fetching prompt-relevant reference images to guide text-to-image models.

Collectively, these studies suggest a shift from purely static indexing toward interactive and tool-mediated evidence acquisition, raising new requirements for evaluation protocols to ensure fair and reproducible comparisons.

D. LIMITATIONS

Given the multimodal nature of the field, future studies may incorporate data types not covered by this taxonomy, implying that the taxonomy itself may evolve. Classification criteria may also shift if post-retrieval stages are defined differently. We adopt the conventional perspective of Lewis et al. [7], which treats vision RAG as retrieval followed by LLM or VLM generation, while acknowledging broader formulations such as Zheng et al. [3] that include image generation.

Our synthesis is constrained by corpus coverage from 2021 to 2025.08, potential selection bias, and subjective primary-label assignment. Variability in metrics and data splits further limits direct comparability across studies. To mitigate these risks, we explicitly disclose our inclusion criteria and labeling rules, and we encourage future

benchmarks to standardize corpora, indices, and reporting practices so that performance gains can be attributed to concrete design choices rather than dataset-specific idiosyncrasies.

VI. CONCLUSION

We have described a comprehensive survey of retrieval-augmented methods in computer vision, covering works published between 2021 and 2025. External knowledge sources were organized into six categories-V-UT, V-OT, V-IM, V-IT, V-SG, and V-DM-based on KB form. For each category, we outlined representative datasets, indexing strategies, and downstream tasks, providing a clear framework for understanding the landscape.

Our analysis showed that despite KB differences, most systems adopted dense embeddings with approximate k -NN search (via FAISS/HNSW) as the default, often supplemented by hybrid sparse-dense indexing, late interaction, or multi-step retrieval. Core downstream tasks-VQA, captioning, and image/video classification-appeared across all categories, underscoring the limitation of using KB type alone to differentiate studies.

Nevertheless, the proposed taxonomy provides an intuitive first step toward understanding how RAG-based vision models leverage different forms of knowledge. Our findings also underscore the need for multidimensional perspectives in future studies, jointly analyzing KB type, retrieval strategies, computational resources, and evaluation protocols. Key challenges include integrating multiple KBs, designing unified evaluation metrics, and establishing reliability-focused methodologies. We expect the framework and discussion in this paper to serve as a foundation for future studies in this domain.

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